TUBULAR & TUBULOINTERSTITIAL DISEASES

- parenchymal diseases is discussed under 2 headings:

- **I. Primary tubular diseases**
- that include tubular injury by ischaemic or toxic agents i.e. acute tubular necrosis.

- **II. Tubulointerstitial diseases**
- that include inflammatory involvement of the tubules and the interstitium i.e. pyelonephritis (acute and chronic)

Acute Tubular Injury/ Necrosis (ATN or ATI)

- It is the most common cause of acute kidney injury
- It is reversable process
- Causes;
  - Ischemia: PAN, malignant hypertension, hypovolemic shock.
  - Direct toxic injury: drugs (gentamicin), heavy metals (mercury), myoglobin (skeletal muscle injuries) and radiation.
  - Acute tubulointerstitial nephritis due to hypersensitivity reaction to drugs (i.e. not bacterial).
  - Disseminated Intravascular Coagulation (DIC).
  - Urinary obstruction by: tumors, BPH, or blood clots (post renal, acute renal failure).

Pathogenesis

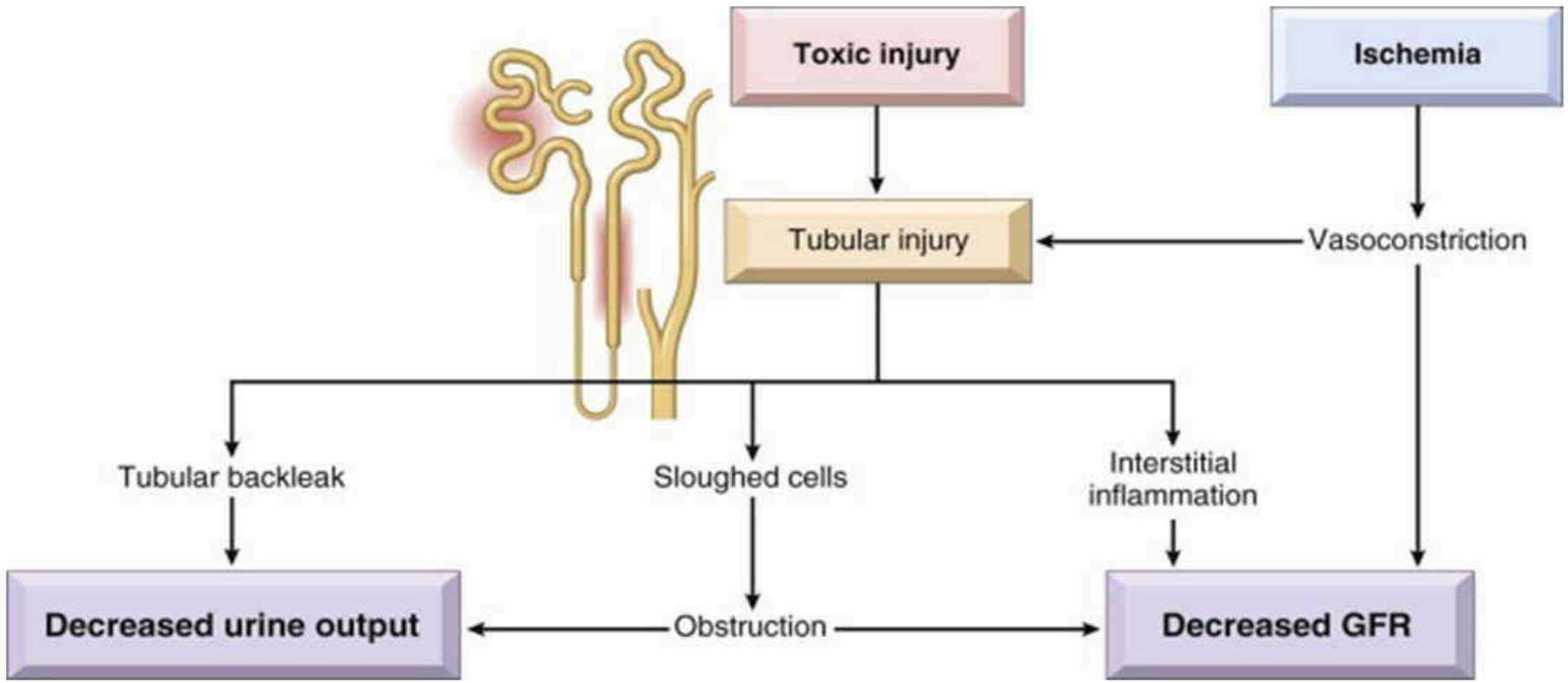
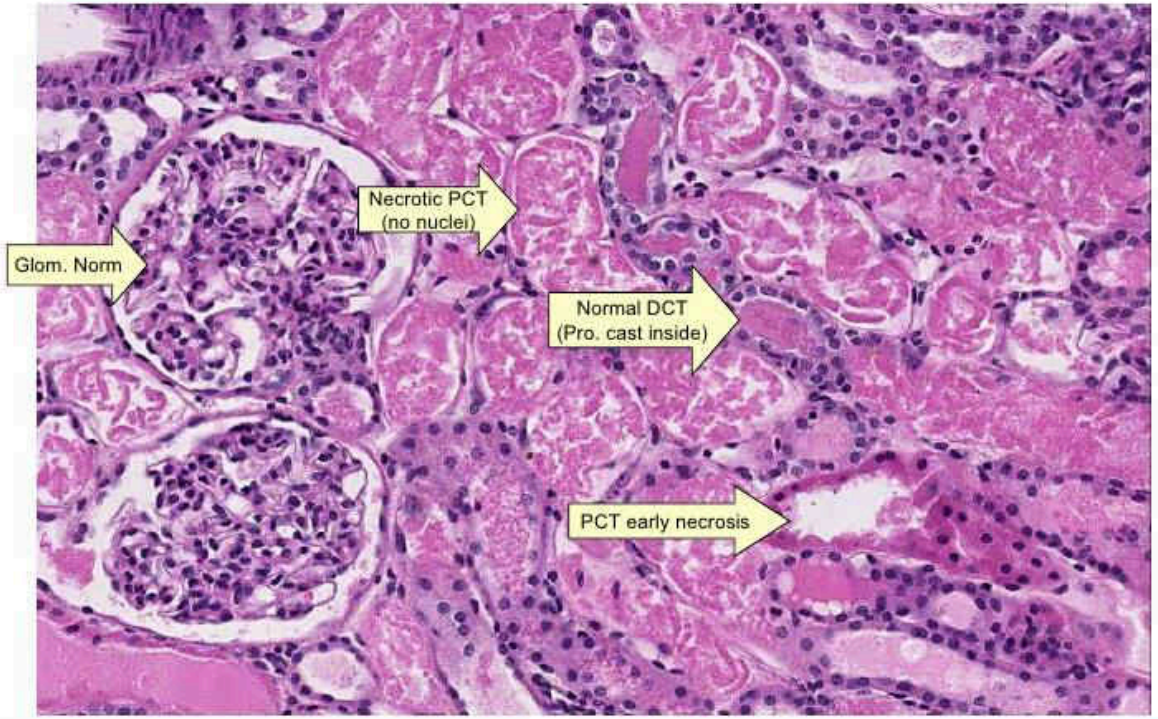
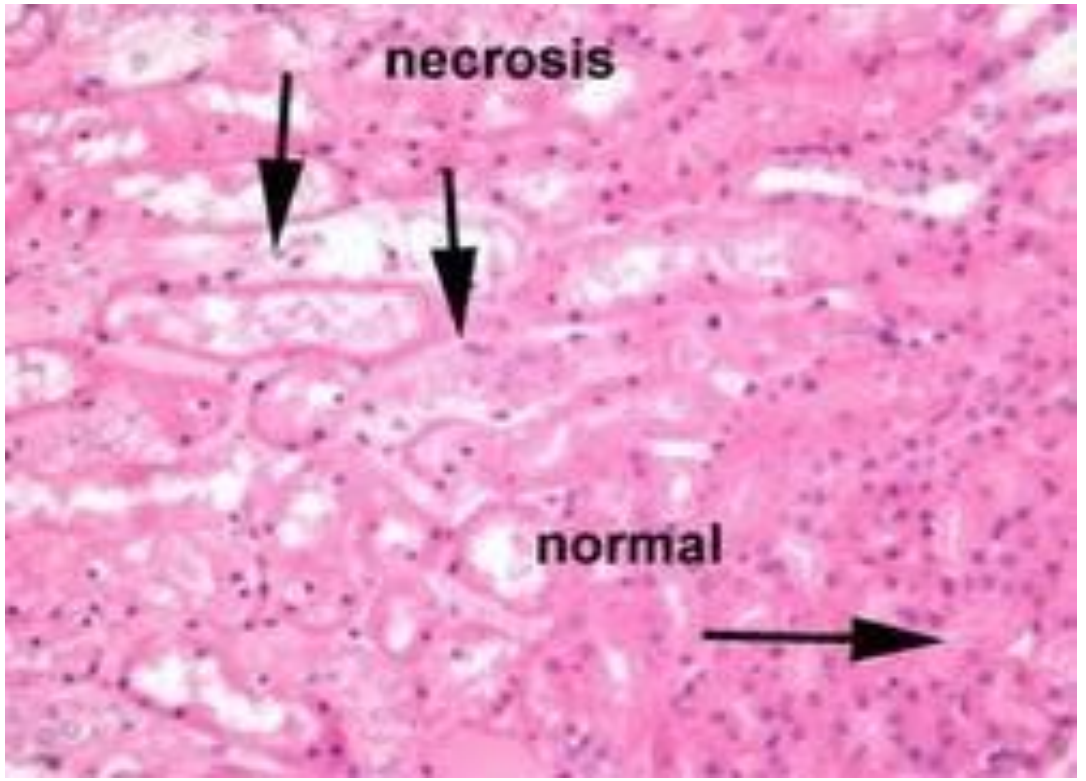


Figure 20-22

Alpers CE, Chang A. Kumar V, et.al., 2015
Robbins and Cotran Pathologic Basis of Disease, 9th Edition

Acute Tubular Necrosis:





acute tubular necrosis. Attenuation of tubule luminal cells with increased eosinophilic staining. Tubule lumens filled with hyaline and fibrin

Tubulointerstitial Nephritis:

- This group of renal diseases involves inflammatory injuries of the tubule & interstitium that are often insidious in onset and are principally manifest by azotemia.
- Acute tubulointerstitial nephritis.
- Chronic tubulointerstial nephritis.

Causes of tubulointerstitial nephritis

Infections
Acute bacterial pyelonephritis
Chronic pyelonephritis (including reflux nephropathy)
Other infections (e.g., viruses, parasites)
Toxins
Drugs
Acute-hypersensitivity interstitial nephritis
Analgesics
Heavy metals
Lead, cadmium
Metabolic Diseases
Urate nephropathy
Nephrocalcinosis (hypercalcemic nephropathy)
Oxalate nephropathy
Physical Factors
Chronic urinary tract obstruction
Neoplasms
Multiple myeloma (light-chain cast nephropathy)
Immunologic Reactions
Transplant rejection
Sjögren syndrome
Sarcoidosis
Vascular Diseases
Miscellaneous
Nephronophthisis
"Idiopathic" interstitial nephritis

Pyelonephritis

- Is one of most common disease of kidney and is defined as inflammation affecting the tubules, interstistium and renal pelvis
- 2 forms

  Acute pyelonephritis

  Chronic pyelonephritis

- Most common cause;

✓85% Gram negative bacteria that are normal habitants in intestinal tract (E.coli, Proteus, Klebsilla, TB……)

✓In immunocompromised patients (polyomavirus, CMV….)

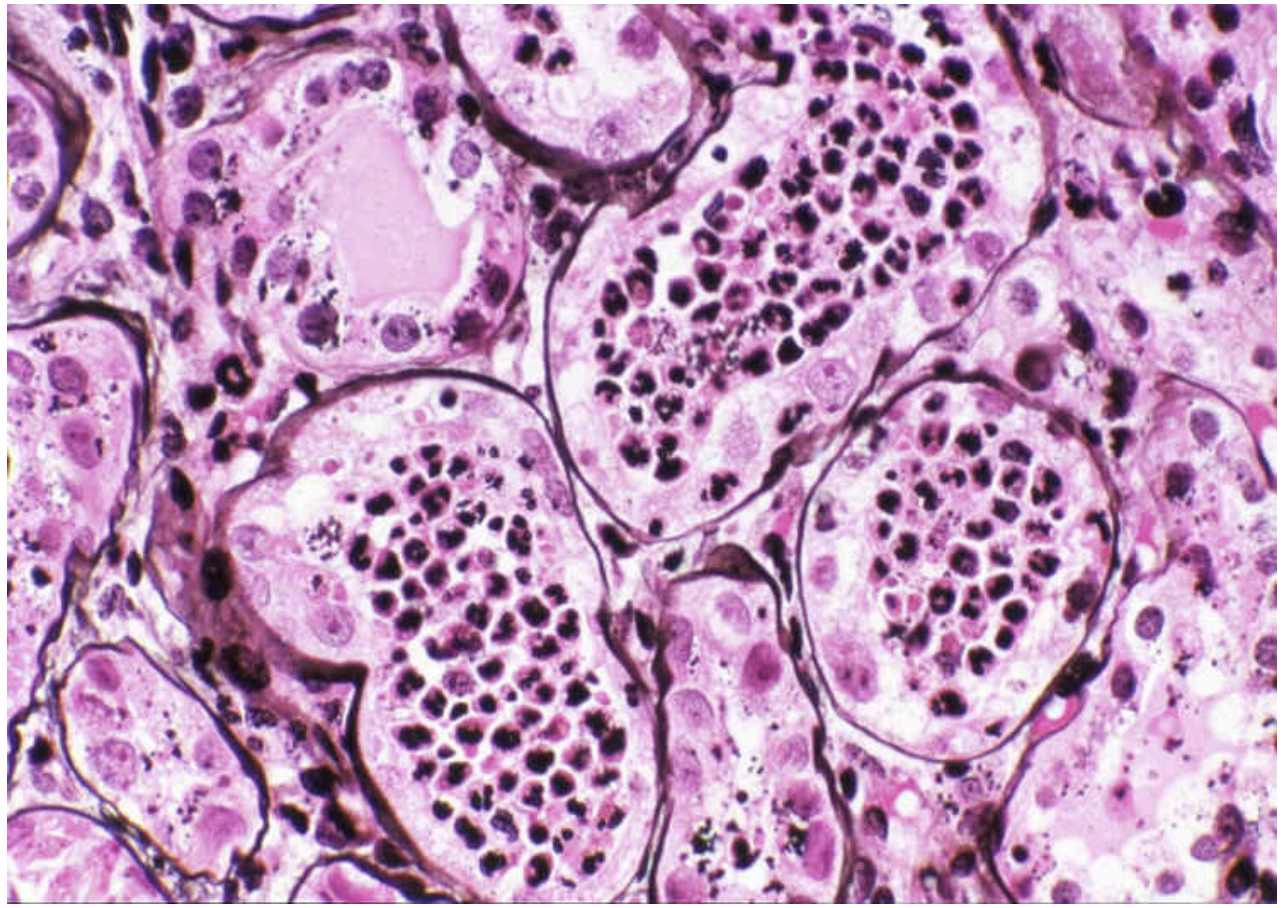
There are 2 way by which bacteria can reach the kidney

- Hematogenous, or ascending (vesicoureteral reflux)

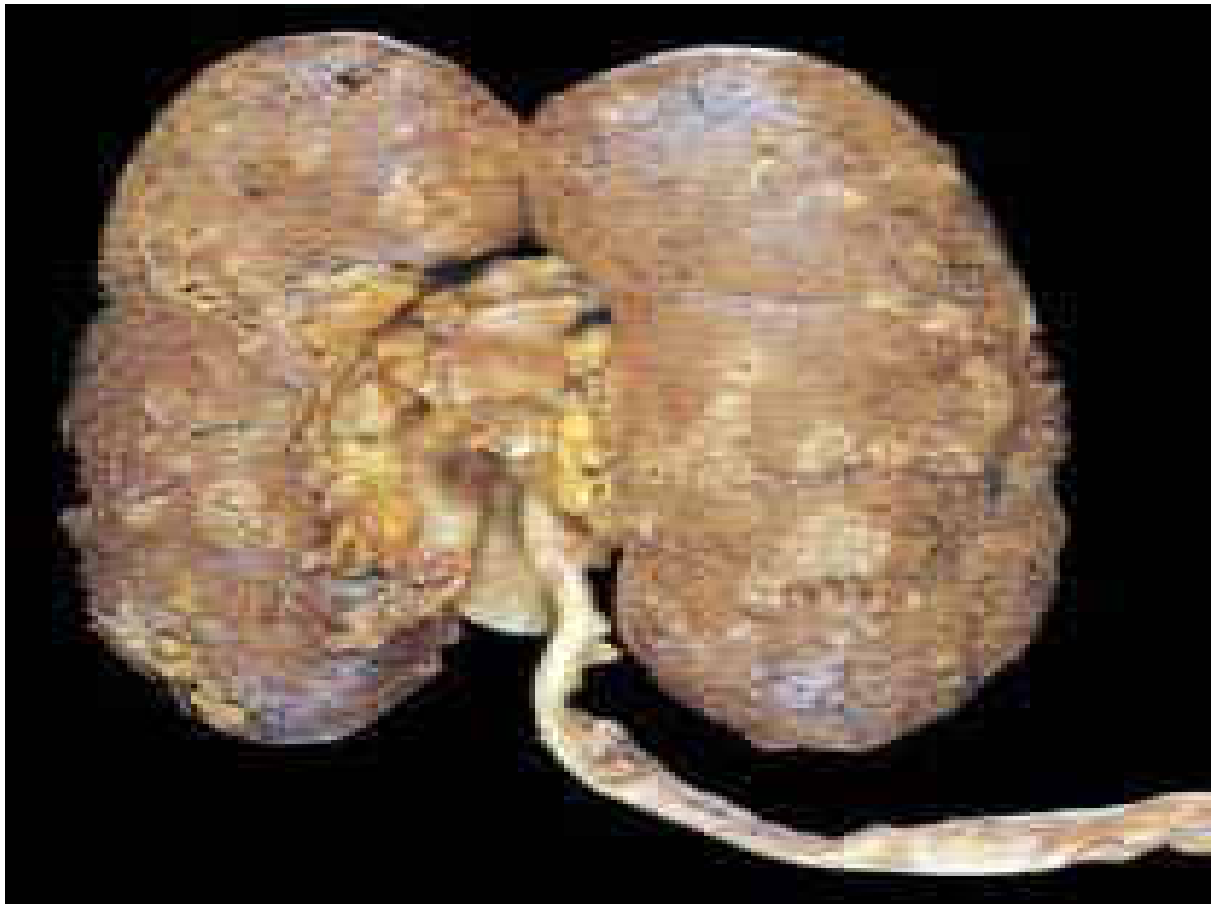
- Onset is sudden with pain, fever, and malaise.
- Bladder and urethral irritation: dysuria, frequency, urgency.
- Urine contains many pus cells (leukocytes): pyuria.
- Pyuria does not differentiate upper from lower UTI; but Leukocyte casts (pus casts) does, because casts are formed only in tubules.
- Diagnosis is established by urine culture.

- **Morphology:**
  - **Gross: cortical surface shows grayish- white areas of inflammation and abscess formation.**
  - **Microscope: Patchy interstitial suppurative inflammation, intratubular aggregates of neutrophils, and tubular necrosis.**

Microscope:



Gross:



complications of acute pyelonephritis:

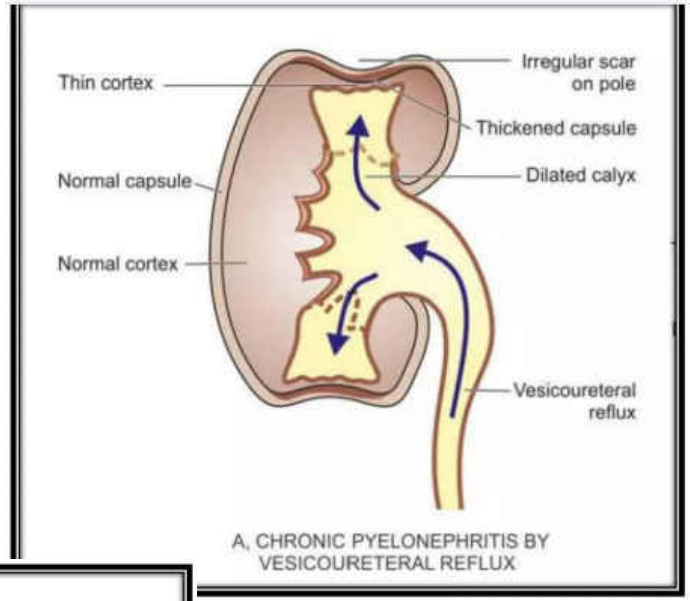
1. **Papillary necrosis:** mainly in diabetics, and in those with urinary tract obstruction.
2. **Pyonephrosis:** when there is total obstruction, the suppurative exudate is unable to drain and thus fills the renal calyces, renal pelvis, and ureter.
3. **Perinephric abscess:** extension of suppurative inflammation through the renal capsule into perinephric tissue

Chronic Pyelonephritis:

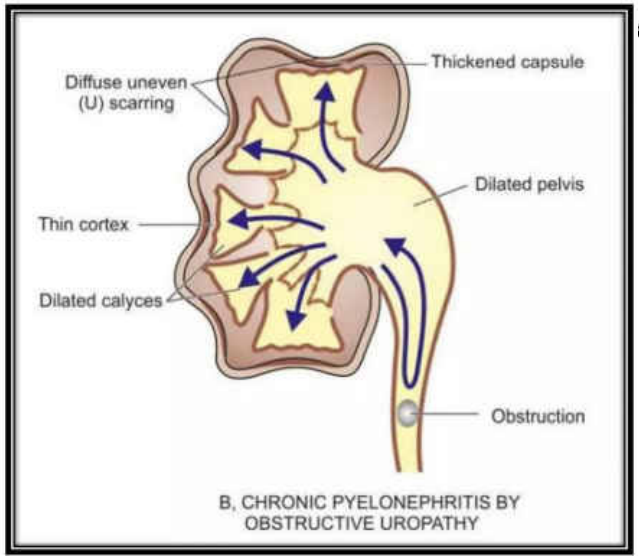
- Chronic tubulointerstitial inflammation and renal scarring associated with involvement of renal calyces and renal pelvis.
- Divided into two forms:

➢ Chronic reflux-associated pyelonephritis (congenital vesicoureteral reflux).

➢ Chronic obstructive pyelonephritis (renal stone).



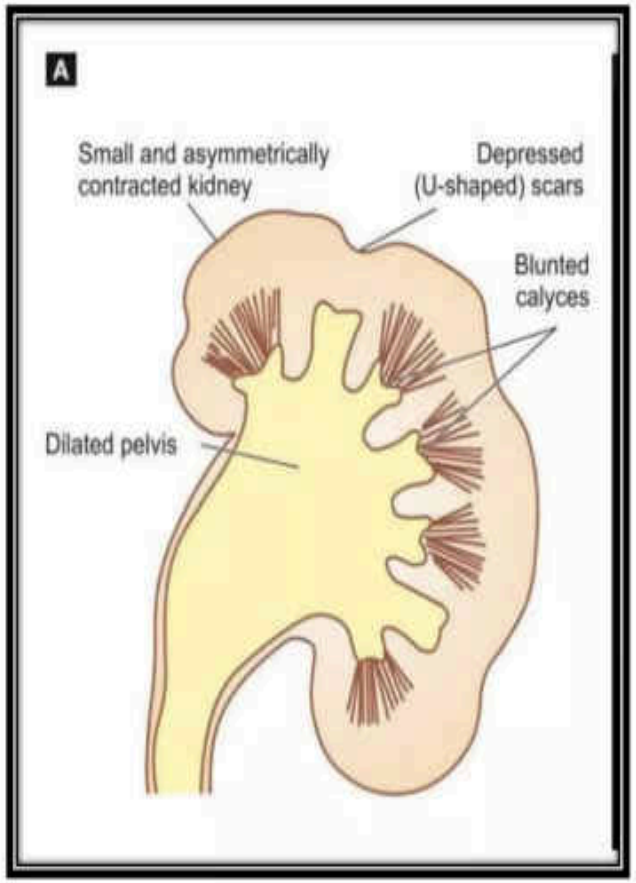
A, CHRONIC PYELONEPHRITIS BY VESICOURETERAL REFLUX



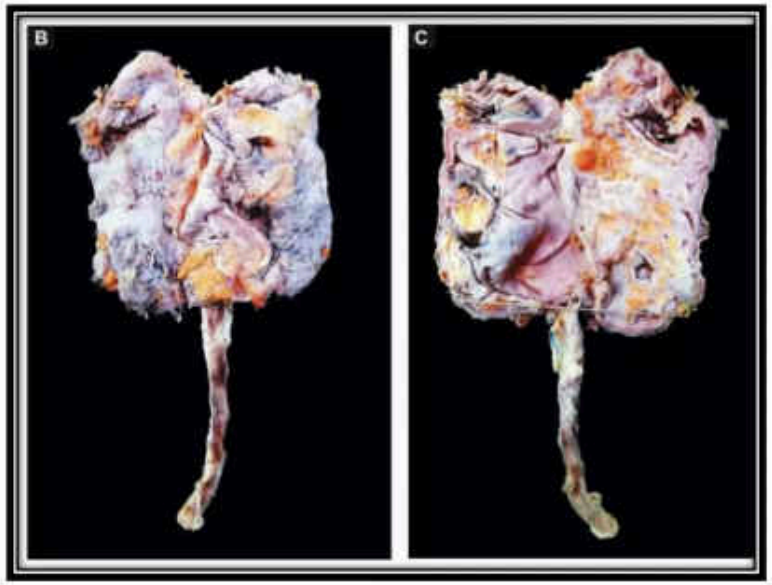
B, CHRONIC PYELONEPHRITIS BY OBSTRUCTIVE UROPATHY

Gross: kidneys are irregularly scarred;

On Cut section: coarse, discrete, corticomedullary scar overlying a dilated, blunted, or deformed calyx



Small contracted kidney in chronic pyelonephritis with calyectasis.



Small contracted kidney in chronic pyelonephritis with calyectasis.

B, External surface of small and contracted kidney. The kidney is small in size and contracted. The capsule is adherent to the cortex and has irregular scars on the surface.

C, Sectioned surface shows dilated pelvi-calyceal system with atrophied and thin peripheral cortex

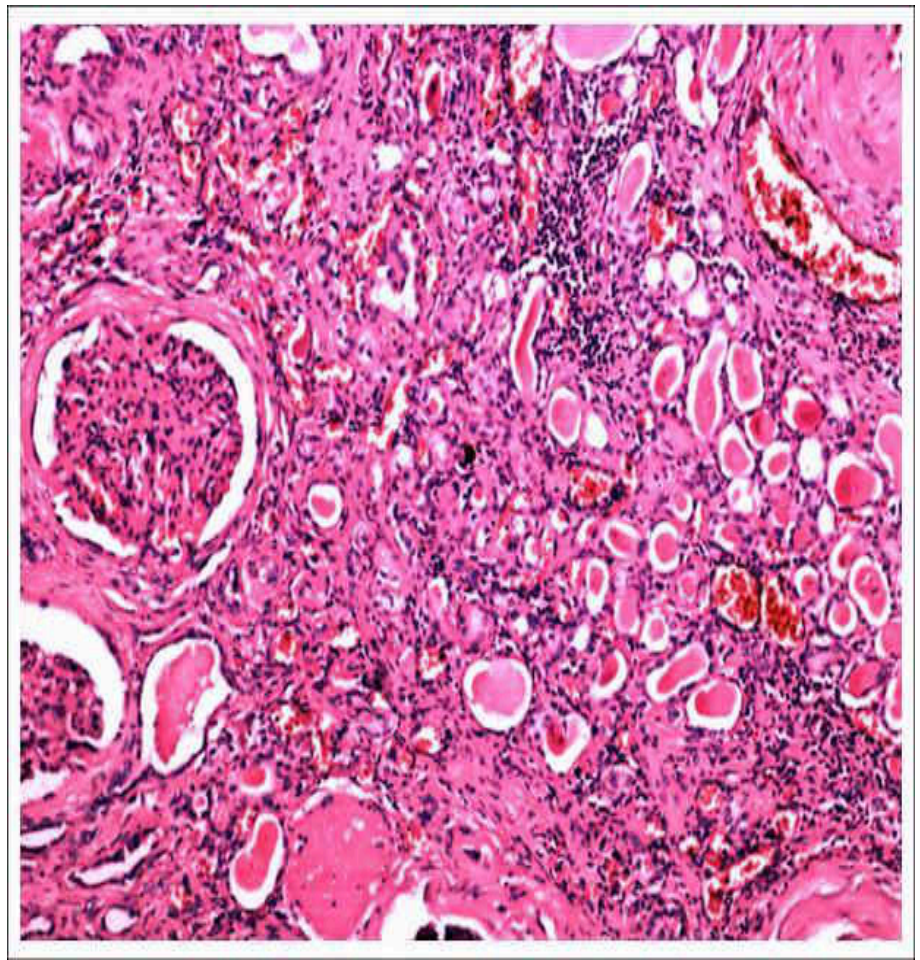
- **M/F**

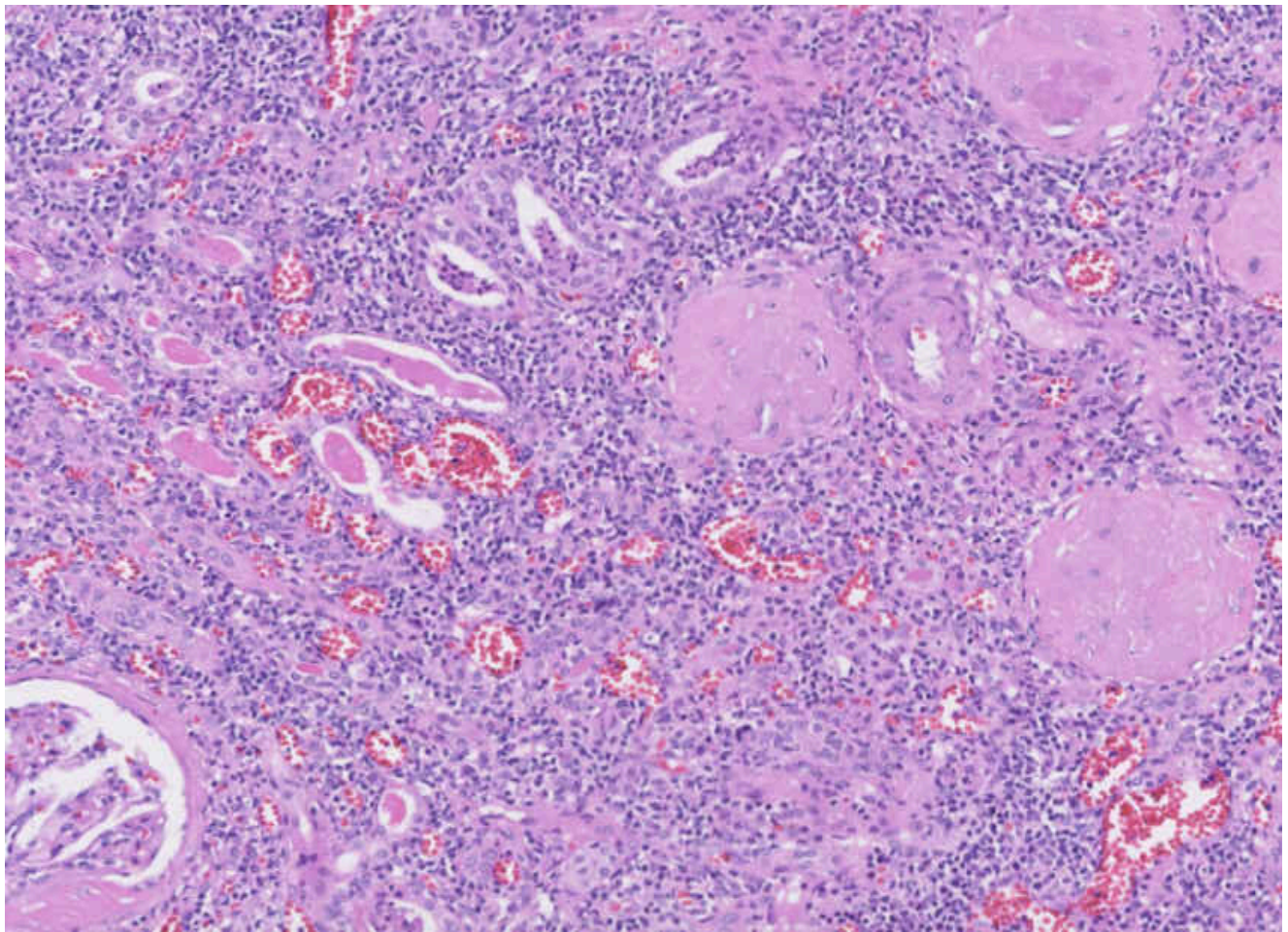
❖ **The tubules show atrophy in some areas, and dilation in others.**

❖ **Tubules lined with flattened epithelium may be filled with colloid casts (thyroidisation).**

❖ **Chronic interstitial inflammation and fibrosis.**

❖ **Glomeruli may appear normal except for periglomerular fibrosis.**



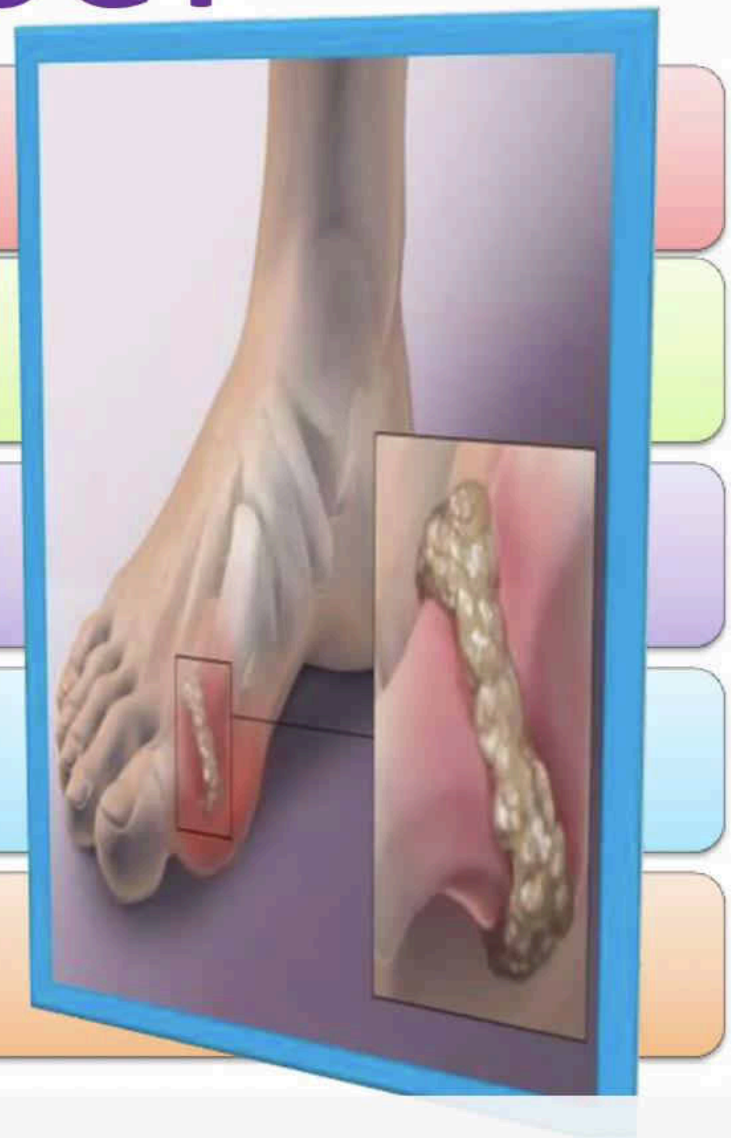


Renal stones

- Nephrolithiasis refers to renal stone disease; urolithiasis refers to the presence of stones in the urinary system. Stones, or calculi, are formed in the urinary tract from the kidney to bladder by the crystallization of substances excreted in the urine
-

ETIOLOGY

- METABOLIC
- LIFESTYLE
- GENETIC FACTORS
- DRUGS
- OTHERS



- Incidence is increased in men.
- Types:

Calcium stones:

- Account for 80% to 85% of urinary stones.
- Consist of calcium oxalate or calcium phosphate, or both.
- Radiopaque.
- Causes:

a. Increased intestinal absorption of calcium.

b. Increased primary renal excretion of calcium.

c. Hypercalcemia by: Hyperparathyroidism, Malignancy (osteolytic metastases or ectopic production of PTH), sarcoidosis, vitamin D intoxication, and milk-alkali syndrome.

Ammonium magnesium phosphate stones:

- Second most common form of urinary stones.
- Formed in alkaline urine caused by ammonia producing (urease-positive) organisms such as *Proteus vulgaris.*
- Radiolucent.
- Can form large staghorn stone.

Uric acid stones:

- Hyperuricemia in approximately half of the patients.
- Hyperuricemia can be secondary to gout; or to increased cellular turnover as in leukemias or myeloproliferative disorders.

- **Cystine stones:** are almost always associated with cystinuria (genetically determined aminoaciduria).

Cystic Diseases of Kidney:

A useful classification of renal cysts is as follows:

1- Polycystic kidney disease:
   a. Autosomal-dominant (adult) polycystic kidney disease.
   b. Autosomal-recessive (infantile) polycystic kidney disease.

2-Multicystic renal dysplasia.

3. Medullary cystic diseases:
   a. Medullary sponge kidney.
   b. Uremic medullary cystic disease (Nephronophthisis).
4. Acquired (dialysis-associated) cystic disease.
5. Simple renal cysts.

Autosomal dominant (adult) polycystic kidney disease **(ADPKD):**

- Its common inherited disorder of kidney
- Manifests clinically between 15 and 30 years of age, although the genetic defect is present at birth.
- AD
- Bilateral
- Both environmental & genetic factors influence the severity.

  blacks (sickle cell trait), in males & hypertension

Extrarenal congenital anomalies (polycystic liver, pancreas, spleen, intracranial berry aneurysms ……etc

Clinical manifestations include:

- Pain
- Hypertension
- Hematuria
- Palpable renal masses
- Progression to renal failure

Genetics and Pathogenesis:

- PKD1 gene encodes a protein named polycystin-1, is located on chromosome 16p13.3 and expressed in tubular epithelial cells.
- Mutation in PDK1 account for about 85%
- <5% develop renal failure by 40yr.s and it increase with age

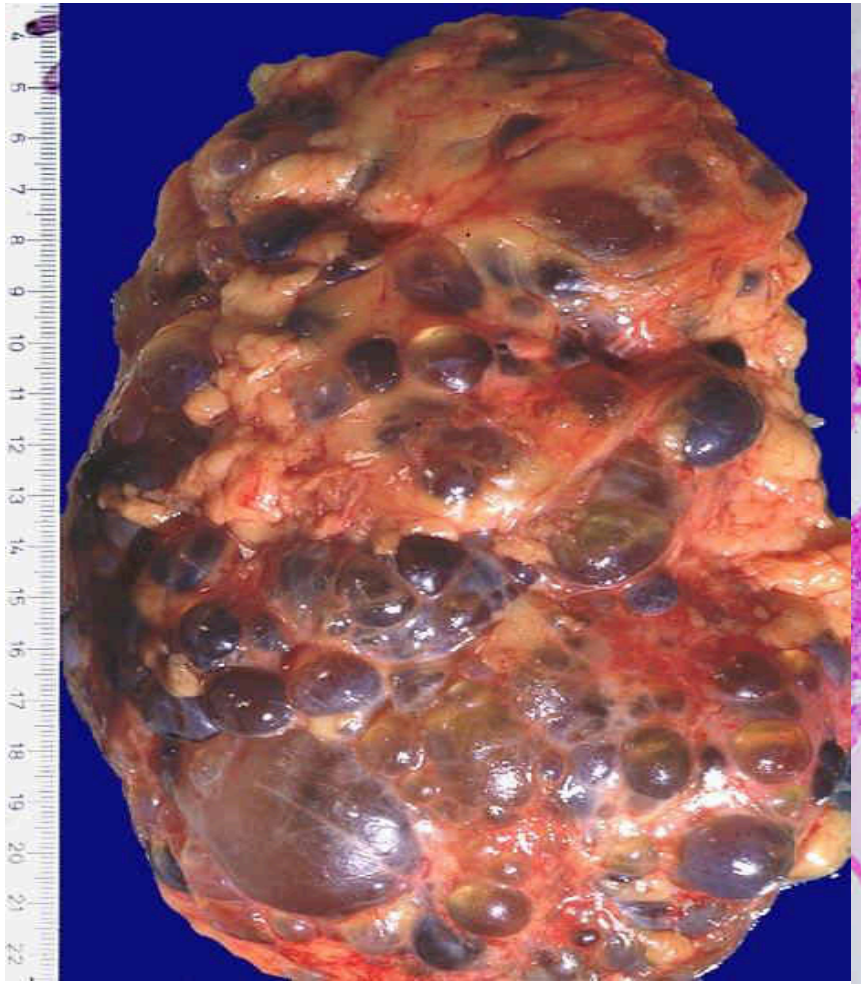
>70% at 60 & 70 years.

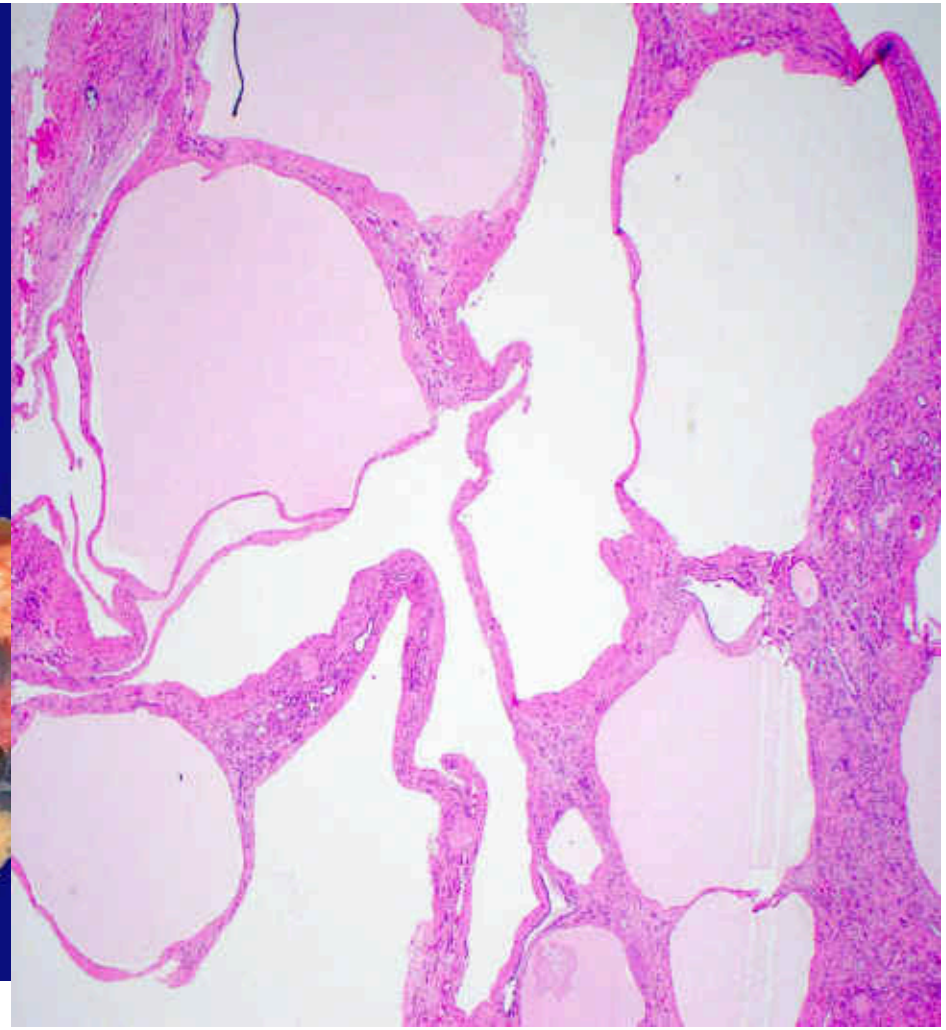
Pathogenesis

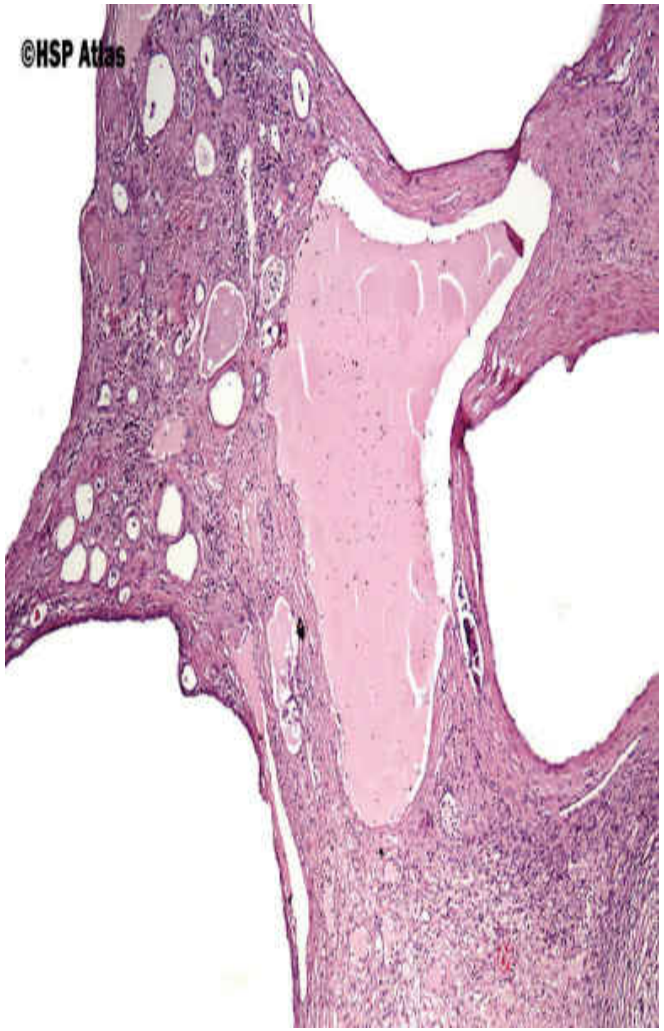
- PKD2 gene product polycystin-2 is located on chromosome 4q21, that expressed in all segments of renal tubules and many extrarenal tissues.
- PDK2< severe than that associated with PKD1 mutation i.e. RF <5% develop at 50yr.s but this rise to 15% at 60 & 45% at 70yr.
- Patients with PKD2 tend to have older age at onset and later development RF.

- **Gross : kidneys are bilaterally enlarged, mass of cysts up to 3 to 4 cm in diameter.**

- **Microscope:**
  - **Cysts filled with a clear serous fluid, or with turbid red to brown or hemorrhagic fluid.**
  - **The cysts arise from the tubules and therefore have variable lining epithelia.**







Autosomal Recessive (childhood) polycystic kidney disease

- Rare, multiple bilateral cysts at birth.
- PKHD1 gene on chromosome 6p21-p23 encodes protein fibrocystin.
- The gene is expressed also in liver and pancreas.
- The function of fibrocystin is unknown but it may be a cell surface receptor with a role in collecting duct and biliary differentiation.
- Patients may develop Congenital Hepatic Fibrosis (CHF): Periportal fibrosis and proliferation of biliary ductules.

- **Gross:**
  - **Kidneys are enlarged with smooth external surface.**
  - **On cut section: numerous small cysts present at right angles to cortical surface.**



- **Microscope:**
  - **Cylindrical or saccular dilations.**
  - **The cysts have a uniform lining of cuboidal cells**

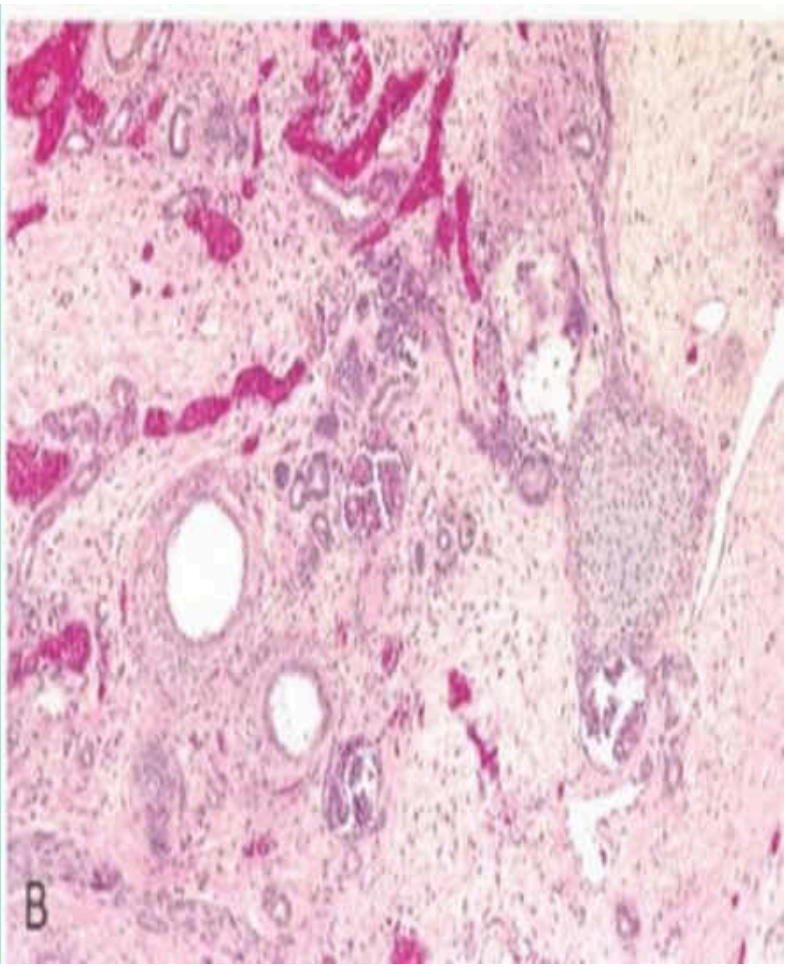


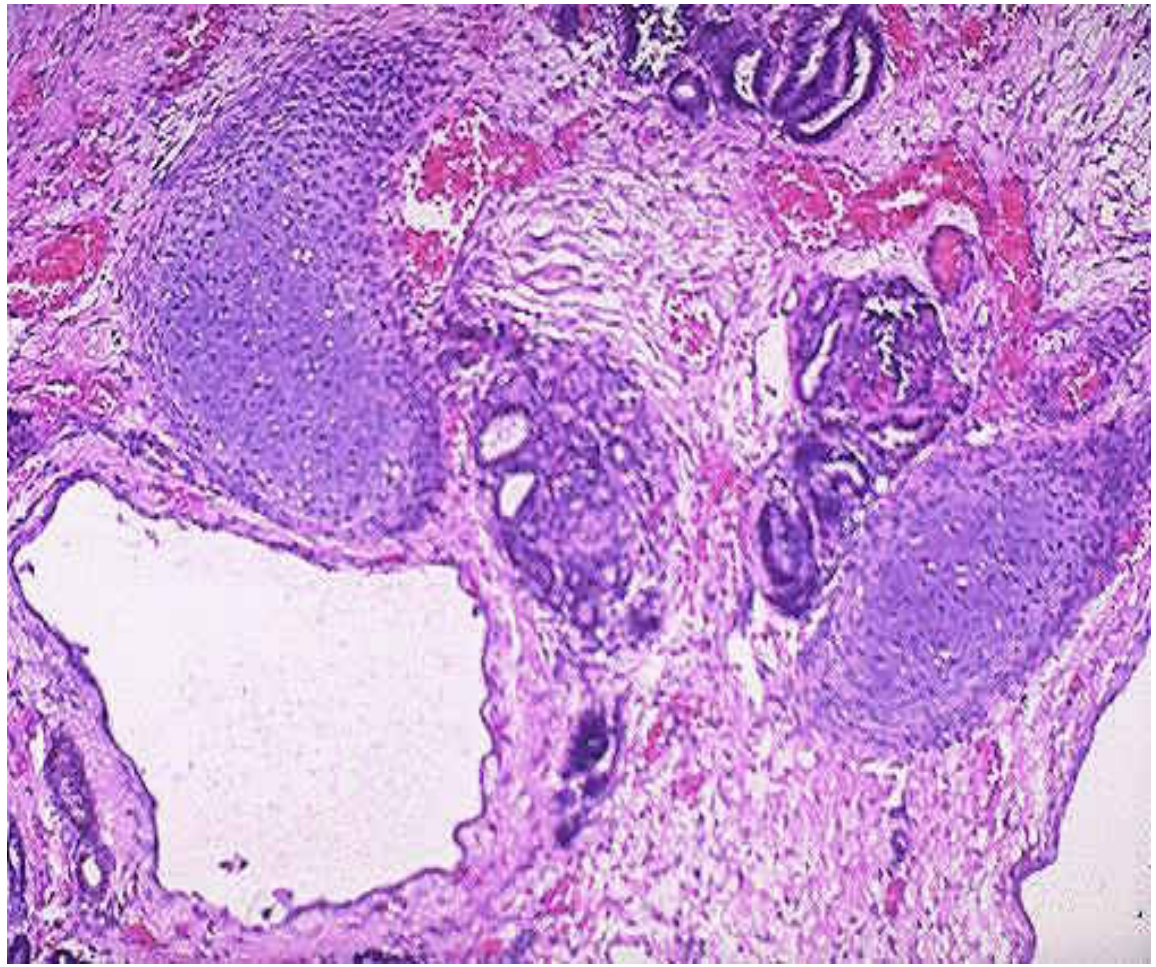
Multicystic renal dysplasia

- **Sporadic.**
- **Due to abnormality in metanephric differentiation.**
- **Can be unilateral or bilateral.**
- **Gross: kidney is enlarged, irregular, and multicystic.**
- **Microscope: cysts are lined by flattened epithelium. the characteristic histologic feature is the presence of islands of undifferentiated mesenchyme, or cartilage and immature collecting ducts**

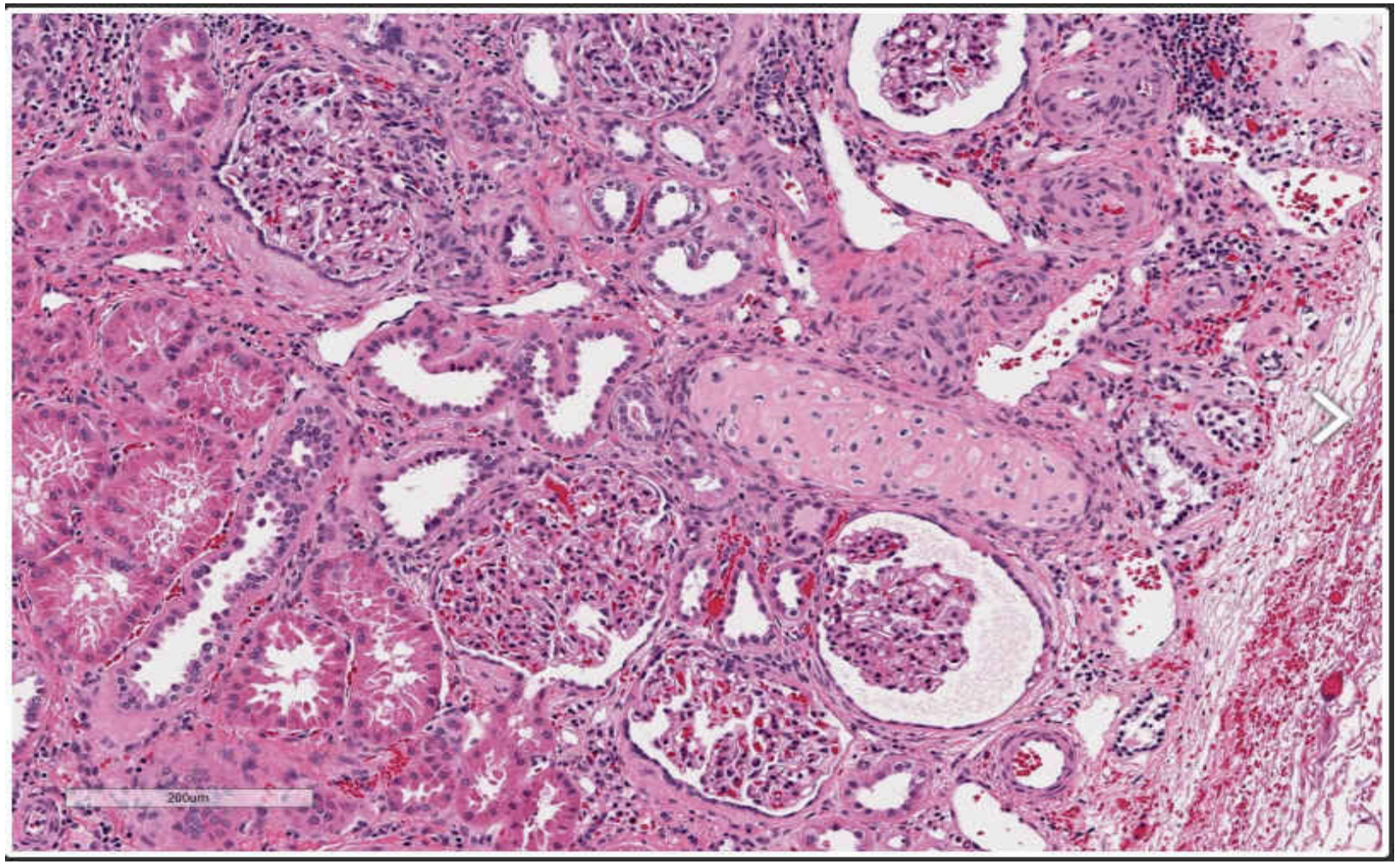
Morphology
Gross & microscopic finding







Multicystic renal dysplasia

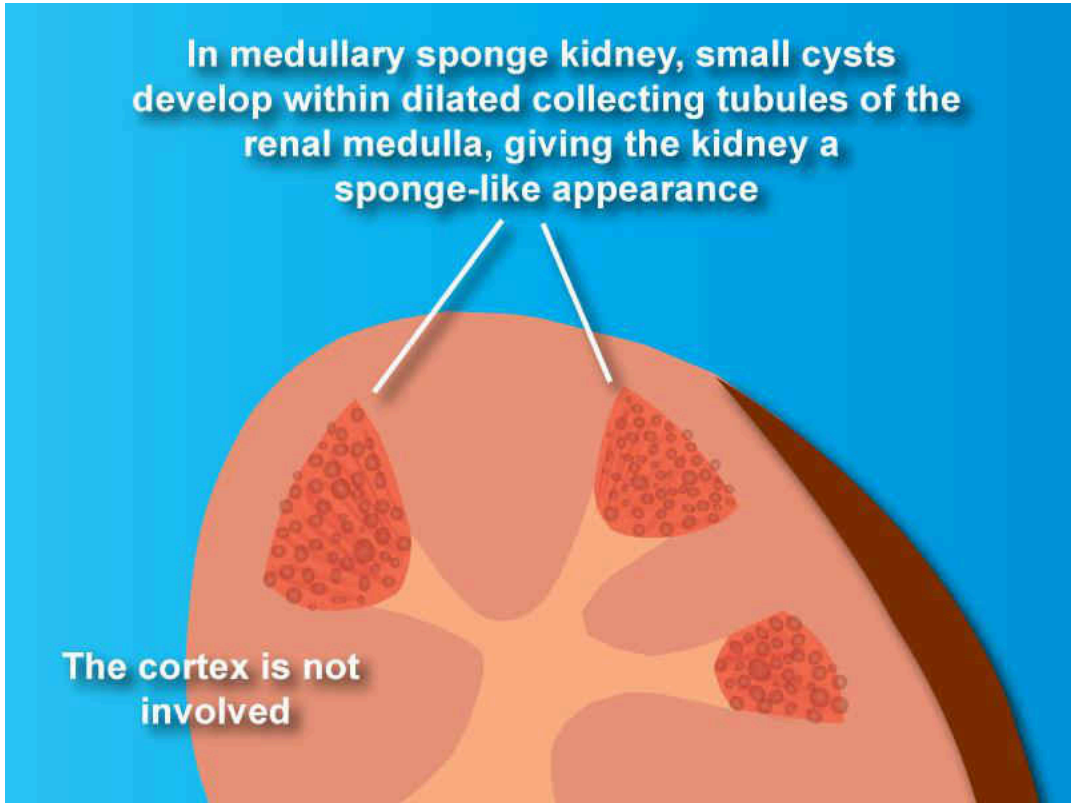


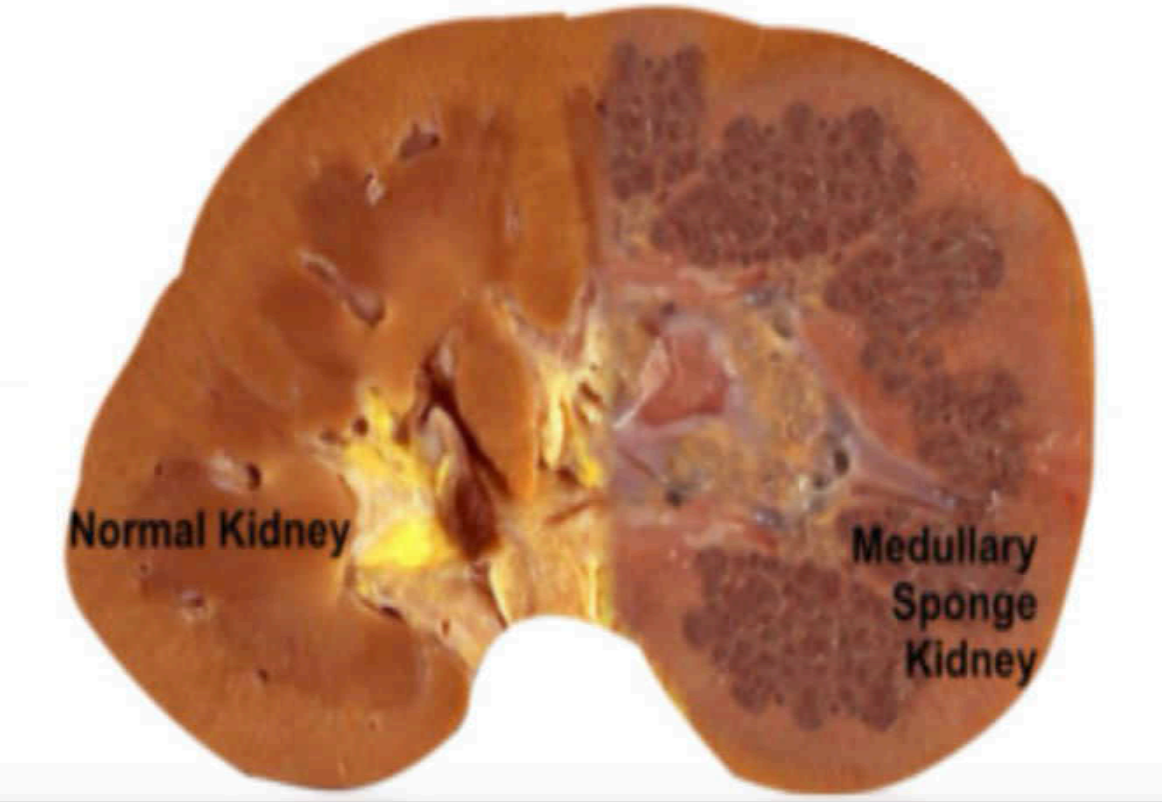
Cystic diseases of renal medulla:

Medullary Sponge Kidney:

- Multiple small medullary cysts usually without renal failure.
- Renal function normal
- Occurs in adults and is usually discovered radiographically as an incidental finding, or may be complicated by infection.
- The pathogenesis is unknown.
- Grossly; papillary ducts in medulla are dilated
- Microscope: the cysts are lined by cuboidal epithelium or occasionally by transitional epithelium.

Medullary sponge kidney





Nephronophthisis

- Cysts in medulla usually at corticomedullary junction.
- The cortical tubulointerstitial damage is the cause of renal insufficiency.
- Progressive renal disorder
- 3 complex variant are recognized;
- 1- sporadic non familial
- 2-Familial juvenile nephronophthisis (most common)
- 3-Renal-retinal dysplasia……. Accompany ocular lesion

- M/F The cysts are lined by flattened or cuboidal epithelium.
- In general glomeruli are preserved.

Gross finding; Small cysts in medulla at cortico-medullary junction with chronic tubulointerstitial nephritis.



Acquired (dialysis-associated) cystic disease:

- Associated with long-term dialysis therapy.
- Multiple cysts, size 0.1-4 cm in diameter, glomerular and tubular atrophy and scarring are characteristic.
- The cysts contain clear fluid and lined by hyperplastic or flattened tubular epithelium.
- Most are asymptomatic, but may cause hematuria.
- There is up to 100-fold increase risk of renal cell carcinoma .

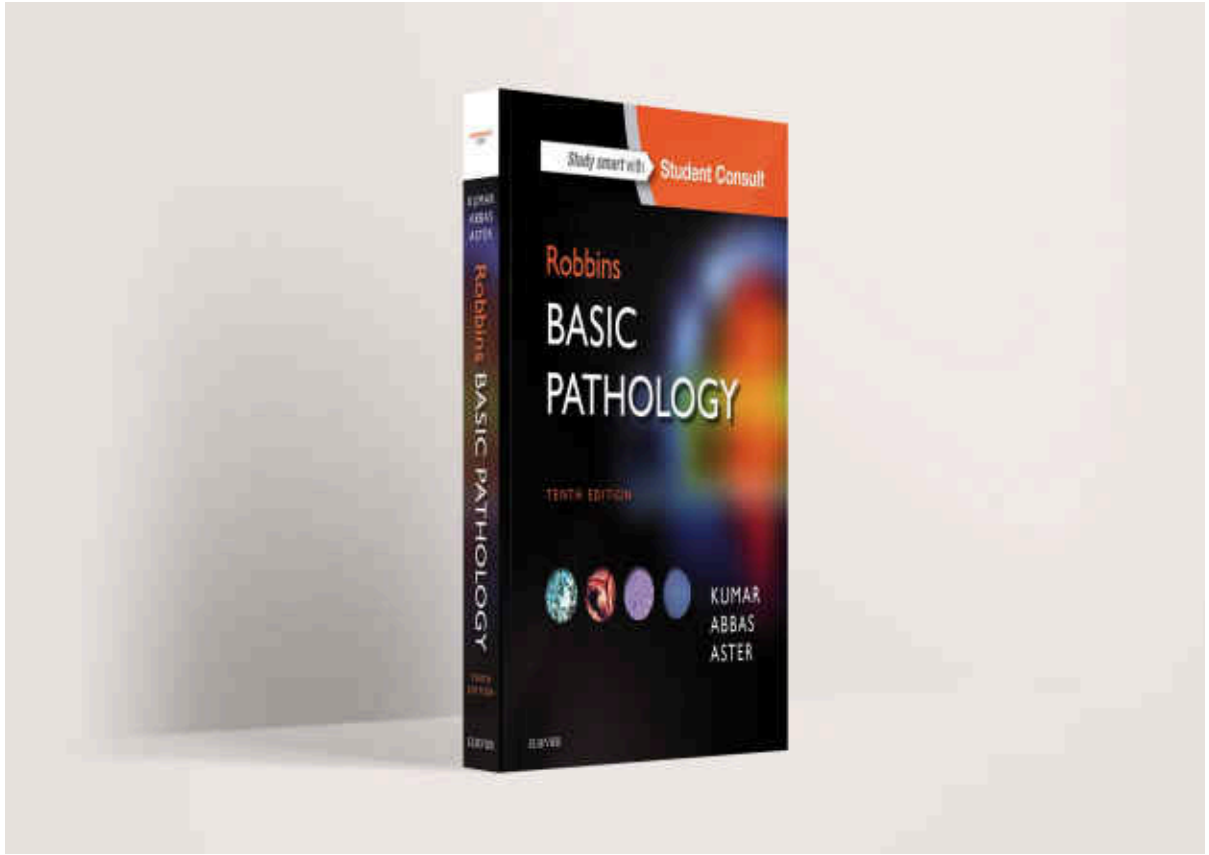
Simple renal cysts:

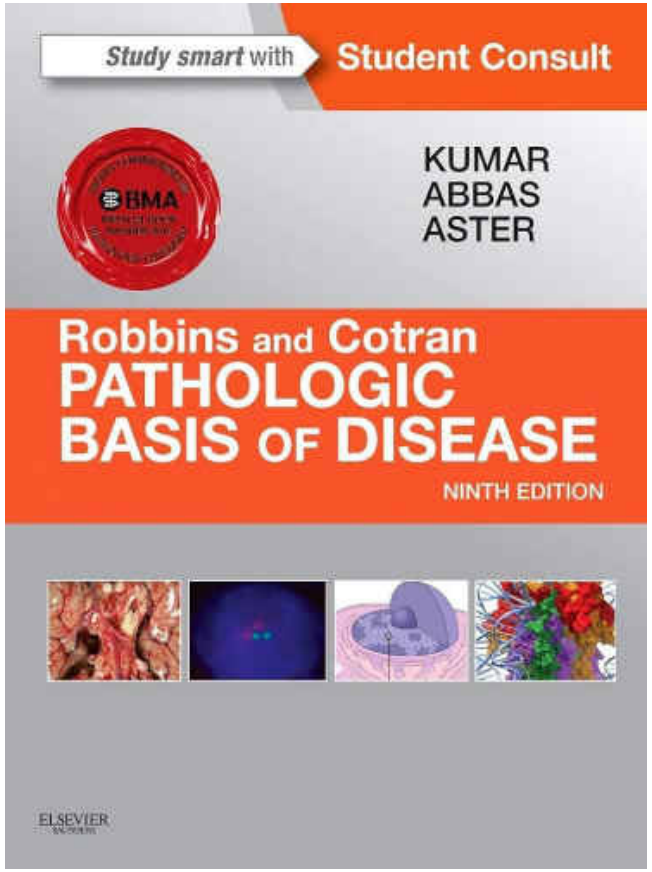
- Most common renal cyst.
- Single or multiple cortical cysts that occur in adults.
- 1 to 5 cm but may reach 10 cm or more in size.
- It is often asymptomatic, but hemorrhage into them may cause sudden distention and pain.
- They are translucent lined by glistening smooth membrane, and filled with clear fluid.
- Microscope: lined by single layer of cuboidal or flattened epithelium.
- The main importance lies in their differentiation from kidney tumors (in contrast to renal tumors they have smooth contours, avascular, and give fluid rather than solid signals on ultrasonography).

Simple renal Cysts



Reference


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